

Java Practice Question Bank

Basics • Operators • Control Statements • Arrays • Method Declaration

Section A — Theory Questions

- Q1.** What is Java? List any three key features of Java (e.g., platform independence, object-oriented, robustness).
- Q2.** Differentiate between JDK, JRE, and JVM.
- Q3.** What is the difference between primitive data types and non-primitive (reference) data types in Java? Give two examples of each.
- Q4.** Explain the difference between `=` (assignment operator) and `==` (equality operator) in Java.
- Q5.** What is type casting in Java? Differentiate between implicit (widening) and explicit (narrowing) type casting with an example.
- Q6.** Differentiate between **while** loop and **do-while** loop. In what situation would a do-while loop be preferred?
- Q7.** What is the difference between **break** and **continue** statements in Java?
- Q8.** What is an array in Java? Why are arrays called a 'reference type' even though they can store primitive values?
- Q9.** What is the default value stored in an array element for (a) int type and (b) String type, when the array is created but not initialized?
- Q10.** Differentiate between a 1-D array and a 2-D array in Java with one real-life example use case for each.
- Q11.** What is a method in Java? What are the components of a method signature?
- Q12.** What is the difference between a **void** method and a method with a **return type**? Give one example of each.
- Q13.** Differentiate between **actual parameters** and **formal parameters** in the context of a method call.
- Q14.** What is the purpose of the **return** keyword? Can a method have multiple return statements? Explain with reasoning.
- Q15.** What is the difference between call by value and call by reference? Which approach does Java use when passing (a) primitive types and (b) arrays to a method?

Section B — Arrays (Declaration & Initialization as Assignment Statements)

For each part below, write the single Java statement(s) that correctly perform the described declaration / assignment. Treat array declaration and initialization itself as an assignment task.

- Q16.** Write a statement to declare an integer array named *marks* of size 5 (without initializing values).
- Q17.** Write a statement to declare and initialize a String array named *days* with values: "Mon", "Tue", "Wed".
- Q18.** Write a statement to declare a 2-D integer array named *matrix* with 3 rows and 4 columns.
- Q19.** Given `int[] arr = new int[5];` write assignment statements to set the first and last elements to 10 and 50 respectively.
- Q20.** Write a statement that creates a double array *prices* directly from the literal values 99.5, 45.0, 120.75.
- Q21.** Write a statement to find and assign the length of an array named *data* to an integer variable *len*.
- Q22.** Write a statement to copy the value of `arr[2]` into a new variable *third*.

Section C — Method Declaration

- Q23.** Write the general syntax for declaring a method in Java, labelling each part (access modifier, return type, method name, parameter list).
- Q24.** Declare a method named *square* that accepts an integer and returns its square (just the declaration line, no body needed).
- Q25.** Declare a void method named *greet* that takes a String parameter *name* and prints a greeting.
- Q26.** Declare a method named *isEven* that accepts an integer and returns a boolean.
- Q27.** Declare a method named *sumArray* that accepts an integer array as a parameter and returns the total sum as an int.
- Q28.** What will happen if a method declared with a return type (e.g., *int*) does not contain a **return** statement in any possible execution path? Explain.

Section D — Coding / Programming Questions

- Q29.** Write a Java program to take two numbers as input and swap their values using a third variable, then print the results.
- Q30.** Write a Java program to check whether a given number is prime or not using a loop.
- Q31.** Write a Java program to find the largest and smallest element in an integer array of 6 elements.
- Q32.** Write a Java program to calculate the sum and average of all elements in an integer array.
- Q33.** Write a Java program to reverse the elements of an array (without using a built-in reverse method) and print the reversed array.
- Q34.** Write a Java program that uses a 2-D array to store a 3x3 matrix and prints the sum of its diagonal elements.

Q35. Write a Java program containing a method *factorial(int n)* that returns the factorial of n, and call it from *main()* for n = 5.

Q36. Write a Java program with a method *maxOfThree(int a, int b, int c)* that returns the largest of three numbers, and demonstrate it in *main()*.

Q37. Write a Java program with a void method *printTable(int n)* that prints the multiplication table of n from 1 to 10.

Q38. Write a Java program that declares an array of 5 integers, passes the array to a method *countPositives(int[] arr)*, and returns the count of positive numbers in the array.

End of Question Paper — refer to the companion 'Java_Solutions.pdf' for approach and full solutions.